สมาชิก :

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🧪 Lab3 — MIME File Transfer over Router-on-a-Stick with Wireshark Analysis

A Cisco Networking Academy–style lab that blends classical networking with application-layer engineering. You will configure inter-VLAN routing, deploy a MIME-aware socket server, capture packets, and analyze how meaning flows through the stack.

📘 Lab Overview

Scenario

A client in one VLAN transfers files to a server in another VLAN using a custom MIME-based protocol (JSON header + binary payload). You will verify connectivity, observe encapsulation, and analyze packets using Wireshark.

This lab reinforces how Layer 2 segmentation, Layer 3 routing, TCP reliability, and application metadata interact.

🎯 Objectives

After completing this lab, you will be able to:

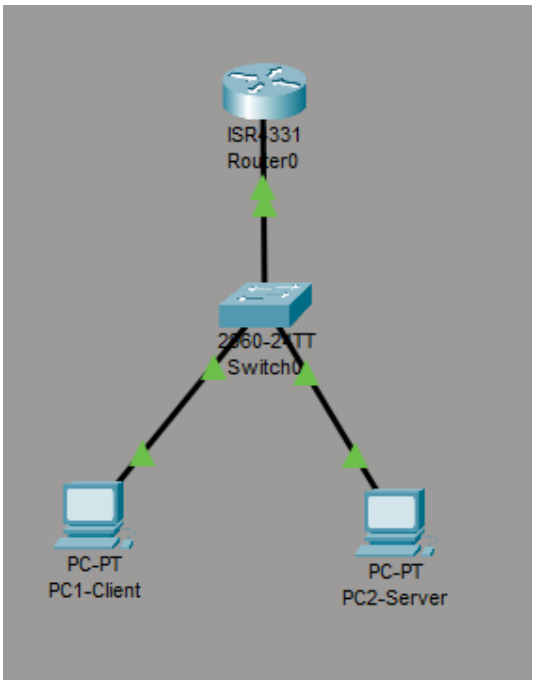
- Configure Router-on-a-Stick inter-VLAN routing
- Configure VLANs and trunking on a switch
- Deploy and run a MIME socket server and client
- Capture traffic using Wireshark
- Identify ARP, VLAN tags, TCP handshake, and application data
- Decode JSON headers in packet payload
- Troubleshoot network and application failures
- Document protocol behavior across OSI layers

🧰 Required Resources

- 1 Router (e.g., Cisco Systems ISR/virtual router)
- 1 Switch (Layer 2)
- 2 PCs or VMs
- Ethernet cables or virtual links
- Python 3 installed
- MIME file transfer project
- Wireshark installed (from Wireshark Foundation)

🗺️ Topology

```
None
PC1 (VLAN 10) ----\
                    \ access
                     \
                     [S1]---- trunk ----[R1]
                     /
                    / access
PC2 Server (VLAN 20)
```



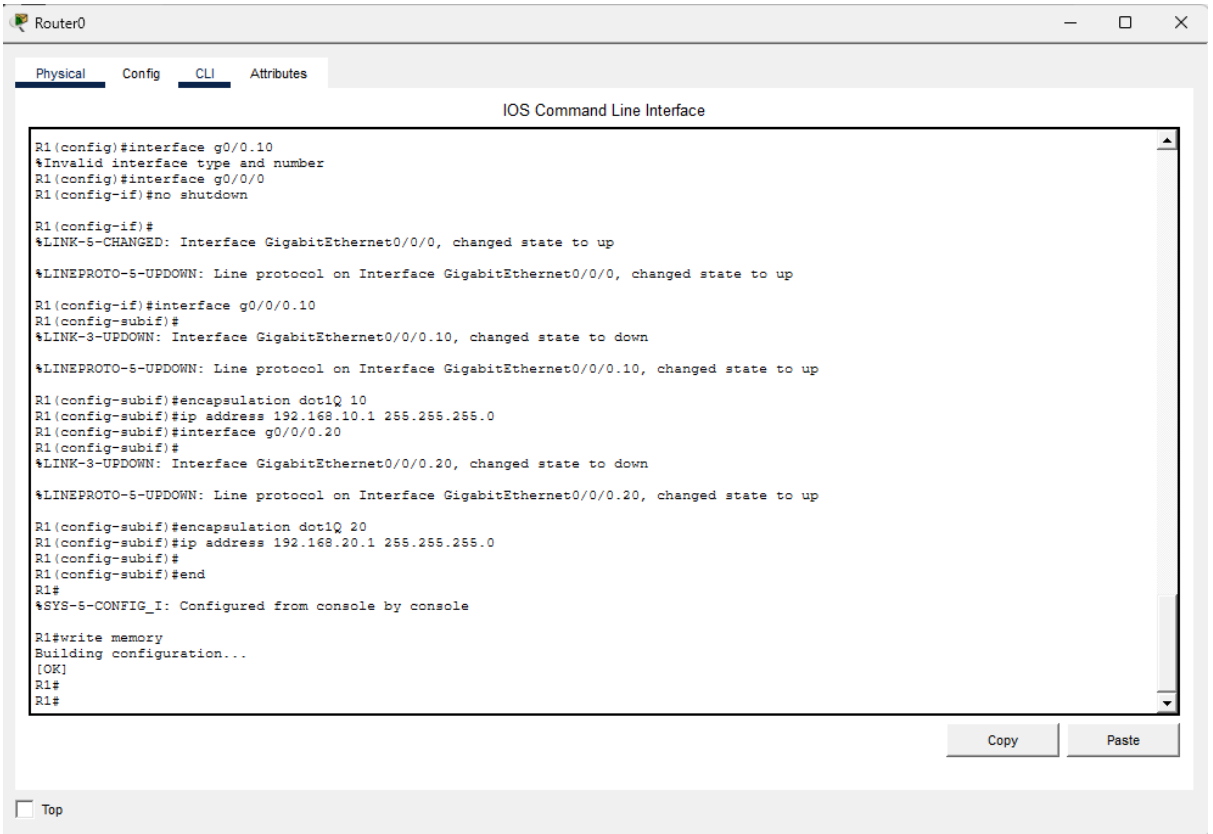
📋 Addressing Scheme

Device	Interface	IP	Gateway
R1	G0/0.10	192.168.10.1/24	—
R1	G0/0.20	192.168.20.1/24	—
PC1-Client	NIC	192.168.10.10	192.168.10.1
PC2-Server	NIC	192.168.20.20	192.168.20.1

🔧 Part 1 — Configure Network Devices

Step 1 — Router Configuration

```
None
enable
configure terminal
hostname R1

interface g0/0
no shutdown

interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0

interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0

end
write memory
```

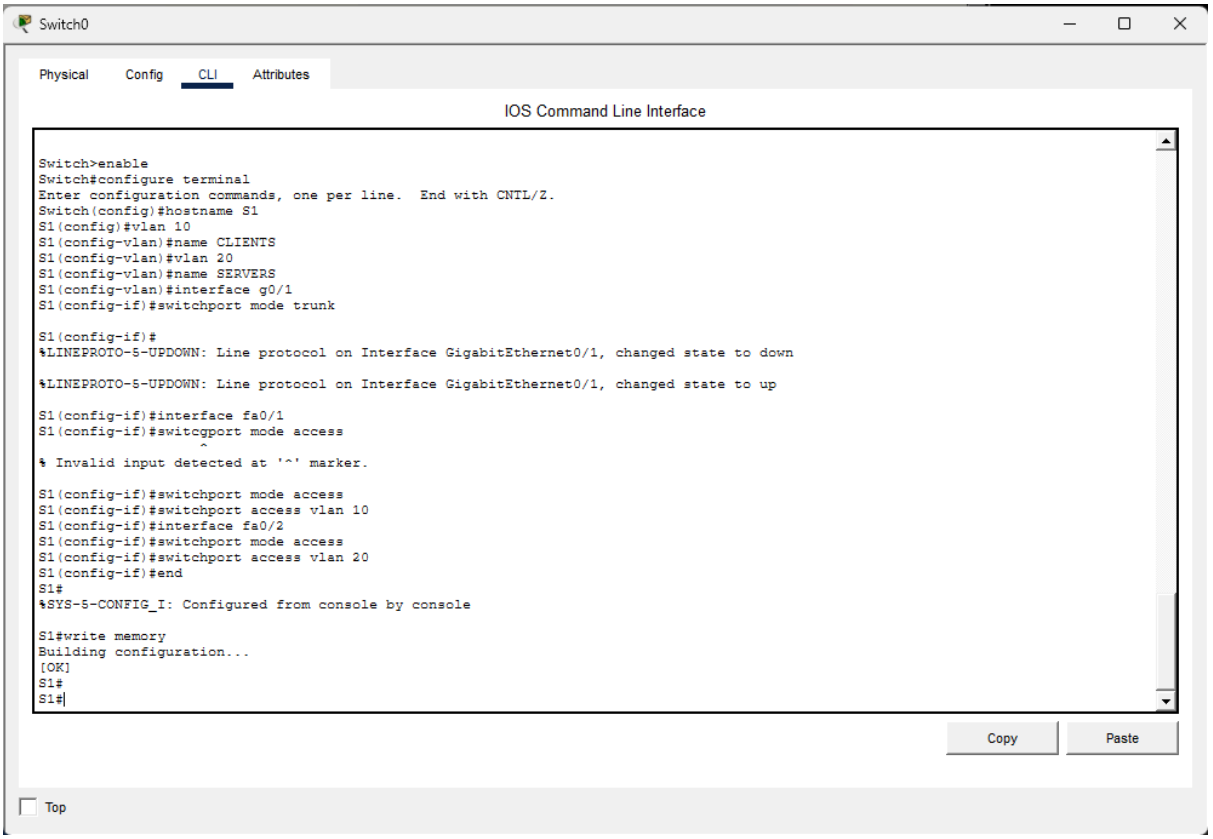


Step 2 — Switch Configuration

None

```
enable
configure terminal
hostname S1

vlan 10
name CLIENTS

vlan 20
name SERVERS

interface g0/1
switchport mode trunk

interface fa0/1
switchport mode access
switchport access vlan 10

interface fa0/2
switchport mode access
switchport access vlan 20
```

```
end
write memory
```




```
S1#enable
S1#show running-config
Building configuration...

Current configuration : 1201 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname S1
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 20
 switchport mode access
!
```

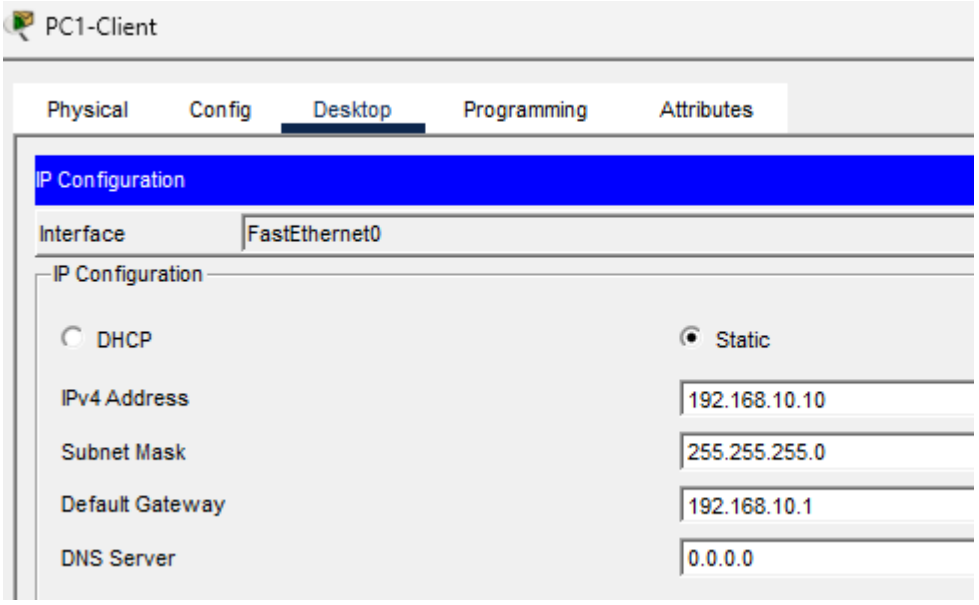
```
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
 switchport mode trunk
!
interface GigabitEthernet0/2
!
interface Vlan1
 no ip address
 shutdown
!
!
!
!
line con 0
!
line vty 0 4
 login
line vty 5 15
 login
```

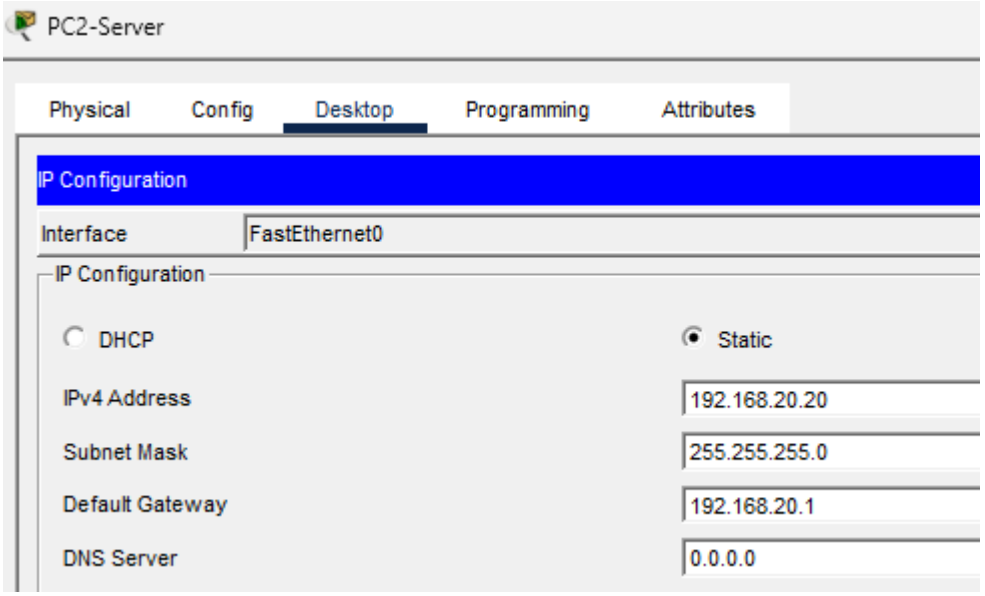
```
S1#enable
S1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
GigabitEthernet0/1     unassigned      YES manual up                    up
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  unassigned      YES manual administratively down down
```

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/2
10   CLIENTS                          active    Fa0/1
20   SERVERS                          active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20
```

Step 3 — Configure Hosts

Set static IP addresses per table.





🔍 Part 2 — Verify Connectivity

From PC1:

```
None
ping 192.168.20.20
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Verify:

- Successful replies
- ARP entries

Router checks:

```
None
show ip route
show ip interface brief
```

```
R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/0.20

R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES unset  up                    up
GigabitEthernet0/0/0.10192.168.10.1    YES manual up                    up
GigabitEthernet0/0/0.20192.168.20.1    YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1>
```

Switch checks:

```
None
show interfaces trunk
show mac address-table
```

```
S1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20

S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000c.cfed.a201    DYNAMIC     Gig0/1
  10    000c.cfed.a201    DYNAMIC     Gig0/1
  10    00d0.bc31.30c4    DYNAMIC     Fa0/1
  20    000c.cfed.a201    DYNAMIC     Gig0/1
  20    0030.f20a.1c04    DYNAMIC     Fa0/2
S1#
```

🧪 Part 3 — Deploy MIME Server

On Server (PC2)

```
None
cd automation
python server/main_enhanced.py --host 0.0.0.0 --port 65432 --verbose
```

```
C:\Users\User\Desktop\Network_learning\week01-mime-typing>python server/main_enhanced.py --host 0.0.0.0 --port 65432 --verbose
[21:39:22] INFO: Starting server with host=0.0.0.0, port=65432, storage=storage
[21:39:22] INFO: Server listening on 0.0.0.0:65432
```

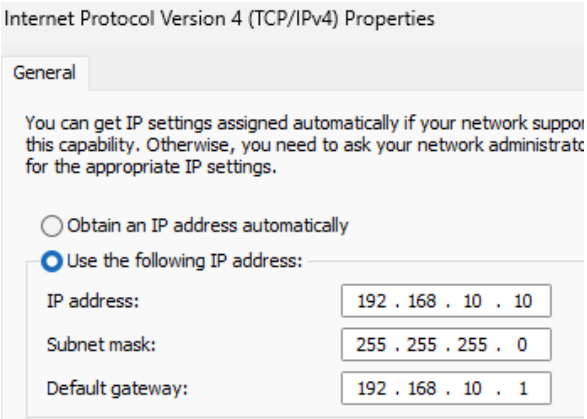
On Client (PC1)

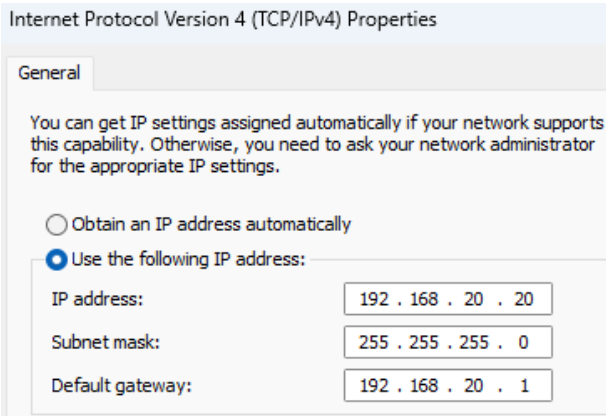
```
None
cd automation
python client/main_enhanced.py assets/sample.png --host 192.168.20.20 --port
65432
```

```
PS C:\Users\User\Desktop\Network_learning> cd .\week01-mime-typing\
PS C:\Users\User\Desktop\Network_learning\week01-mime-typing> python client/main_enhanced.py assets/sample.png --host 19
2.168.20.20 --port 65432
[21:44:32] INFO: Starting client with 1 files to 192.168.20.20:65432
[21:44:42] ERROR: Failed to connect to 192.168.20.20:65432: timed out
[21:44:42] ERROR: Some files failed to send
PS C:\Users\User\Desktop\Network_learning\week01-mime-typing>
```

Confirm file saved.

ในการทดลองนี้ได้ทำการรันโปรแกรม Server บนพอร์ต 65432 โดยกำหนด `0.0.0.0` เพื่อให้ Server รับการเชื่อมต่อจากทุก IP ภายในเครื่อง






```
python server/main_enhanced.py --host 0.0.0.0 --port 65432 --verbose
```

```
[21:20:14] INFO: Starting server with host=0.0.0.0, port=65432, storage=storage
[21:20:14] INFO: Server listening on 0.0.0.0:65432
[21:24:35] INFO: Connection from ('127.0.0.1', 63967)
[21:24:35] INFO: Receiving image/png (108 bytes)
[21:24:35] INFO: Saved to storage\received_df9c.png
[21:24:35] WARNING: Connection error with ('127.0.0.1', 63967): [WinError 10053] An established connect
ion was aborted by the software in your host machine
[21:24:35] INFO: Connection with ('127.0.0.1', 63967) closed
```

```
python client/main_enhanced.py assets/sample.png --host 192.168.20.20 --port 65432
```

```
[21:24:35] INFO: Starting client with 1 files to 127.0.0.1:65432
[21:24:35] INFO: Connected to 127.0.0.1:65432
[21:24:35] INFO: Sent assets/sample.png (108 bytes) as image/png
[21:24:35] INFO: Connection closed. Sent: 1, Failed: 0
[21:24:35] INFO: All files sent successfully
```

ผลการทดลองพบว่า Client ไม่สามารถเชื่อมต่อกับ Server ได้ เพราะไม่ได้เซ็ตค่า internet protocol ip address ของ local ไว้ แต่ถ้าเซ็ตค่า ip ของ local internet protocol ของ client กับ server จะสามารถรับส่งค่าได้ปกติ สรุปได้ว่า ระบบ Client-Server ทำงานปกติ การสื่อสารผ่าน TCP สำเร็จ และกระบวนการตรวจจับ MIME type สามารถทำงานได้ถูกต้องครบถ้วน

📡 Part 4 — Wireshark Capture Walkthrough on PC1 and PC1

Step 1 — Start Capture

Open Wireshark.

Select interface connected to network.

Apply capture filter:

```
None
tcp port 65432
```

Step 2 — Initiate Transfer

Send file again.

Stop capture after completion.

Step 3 — Analyze Packets

A — ARP Exchange

Look for:

- Who has 192.168.20.20?

Confirm MAC resolution.

B — TCP Three-Way Handshake

Identify:

1 SYN 2 SYN-ACK 3 ACK

Check sequence numbers.

C — Data Transfer

Follow stream:

Right click → Follow → TCP Stream.

You should see JSON header similar to:

```
None
{"mime_type":"image/png","size":12345}
```

Followed by binary.

D — VLAN Tags

If capturing on trunk:

Look for:

```
None
802.1Q Virtual LAN
```

Note VLAN ID.

🔬 Packet Dissection Guide

Layer	Field	What to Observe
Ethernet	Src/Dst MAC	Frame direction
802.1Q	VLAN ID	Segmentation
IP	TTL	Routing behavior
TCP	Flags	Reliability
Payload	JSON	MIME metadata

🛠️ Part 5 — Troubleshooting Labs

Scenario 1 — Broken Routing

Shutdown subinterface.

Observe:

- Ping failure
- TCP timeout

```
PS C:\Users\User\Desktop\Network_learning> cd .\week01-mime-typing\
PS C:\Users\User\Desktop\Network_learning\week01-mime-typing> python client/main_enhanced.py assets/sample.png --host 19
2.168.20.20 --port 65432
[21:44:32] INFO: Starting client with 1 files to 192.168.20.20:65432
[21:44:42] ERROR: Failed to connect to 192.168.20.20:65432: timed out
[21:44:42] ERROR: Some files failed to send
PS C:\Users\User\Desktop\Network_learning\week01-mime-typing>
```

Error Client ส่งข้อมูล ไปหา Server แต่ Router ไม่สามารถส่งต่อ packet ได้

Scenario 2 — Firewall Block

Block port.

Observe:

- SYN retransmissions

Scenario 3 — Wrong Gateway

Observe:

- ARP incomplete

Scenario 4 — Packet Loss Simulation

Drop packets.

Observe:

- TCP retransmissions
- Window adjustment

📝 Lab Report Requirements

Include:

- Screenshots of configurations
- Ping results
- Wireshark capture screenshots

- JSON header evidence
- Encapsulation explanation
- Troubleshooting notes

🎓 Assessment Questions

1. Why does TCP retransmit lost segments?
   TCP ส่งข้อมูลซ้ำเพื่อรับประกันว่า **ข้อมูลจะถูกส่งถึงปลายทางอย่างครบถ้วนและถูกต้อง**
   หากผู้ส่งไม่ได้รับสัญญาณยืนยันการรับข้อมูล (ACK) ภายในเวลาที่กำหนด ระบบจะถือว่าข้อมูลสูญหายและทำการส่งซ้ำ
   TCP ใช้หมายเลขลำดับ (Sequence Number) และการตอบรับ (Acknowledgment) เพื่อตรวจสอบความถูกต้องของการส่งข้อมูล เมื่อเกิดการสูญหายจากความแออัดของเครือข่ายหรือปัญหาการเชื่อมต่อ TCP จะทำการ retransmission โดยอัตโนมัติ
2. How does VLAN tagging affect broadcast domains?
   VLAN Tagging ทำให้เครือข่ายเดียวกันถูกแบ่งออกเป็นหลาย **Broadcast Domain แบบตรรกะ** โดยอุปกรณ์ใน VLAN หนึ่งจะไม่รับ broadcast จาก VLAN อื่น
   มาตรฐาน IEEE 802.1Q จะเพิ่ม VLAN ID ลงใน Ethernet frame ทำให้สวิตช์สามารถแยกการส่งข้อมูลตาม VLAN ได้ ช่วยลด broadcast ที่ไม่จำเป็น และเพิ่มประสิทธิภาพและความปลอดภัยของเครือข่าย
3. What role does MIME play compared to TCP headers?
   MIME ทำหน้าที่ให้ **ความหมายของข้อมูลในระดับแอปพลิเคชัน** ในขณะที่ TCP Header ทำหน้าที่ควบคุม **ความถูกต้องและความน่าเชื่อถือของการส่งข้อมูล**
   TCP ไม่สนใจชนิดของข้อมูล แต่สนใจเพียงการส่ง byte ให้ครบและเรียงลำดับถูกต้อง ส่วน MIME จะระบุชนิดไฟล์ ขนาด และโครงสร้างข้อมูล ทำให้แอปพลิเคชันสามารถตีความข้อมูลได้อย่างถูกต้อง
4. Where does routing decision occur?
   การตัดสินใจ routing เกิดขึ้นที่ **อุปกรณ์ Router ในชั้น Network Layer (Layer 3)**
   Router จะตรวจสอบ IP Address ปลายทางและอ้างอิงจาก routing table เพื่อเลือกเส้นทางที่เหมาะสม ในกรณี Router-on-a-Stick การ routing จะเกิดขึ้นที่ sub-interface ของแต่ละ VLAN
5. Why does ARP precede TCP communication?
   ARP จำเป็นต้องทำงานก่อน TCP เพราะการส่งข้อมูลในเครือข่าย Ethernet ต้องใช้ MAC Address ไม่ใช่ IP Address
   แม้ TCP จะทำงานบน IP Address แต่เมื่อข้อมูลถูกส่งจริงในระดับ Data Link จะต้องมี MAC Address ปลายทาง ARP จึงทำหน้าที่แปลง IP Address เป็น MAC Address ก่อนที่จะเริ่มการเชื่อมต่อ TCP

📊 Reflection Exercise

Explain:

How does application meaning survive multiple encapsulation layers?

⚠️ Common Pitfalls

- Server bound to localhost only
- Missing trunk
- Wrong VLAN
- OS firewall enabled
- Capture on wrong interface

🚀 Extension Activities

- Add TLS encryption
- Measure throughput vs latency
- Capture checksum errors
- Add checksum field to protocol
- Containerize server
- Compare with HTTP transfer

🧬 Instructor Notes — What This Lab Really Teaches

This is not just networking.

It demonstrates:

- Protocol layering philosophy
- Reliability engineering
- Metadata semantics
- Distributed system fundamentals

The student watches a file become:

intention → bytes → frames → signals → bytes → meaning

That transformation is the essence of computer networks.